

Making Styrene from Scratch

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CHE396 - Senior Design I

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Project charter:

Title
Making Styrene from Scratch
Goal
To overview the styrene market and design an efficient process to produce 500,000 tons of styrene per year.
In Scope
• Design of ethylbenzene production process • Selection and design of styrene production process • Comparison with competing production processes • Material balances across unit operations
Out of Scope
• Waste treatment • Location selection • Energy and utilities analysis • Catalyst selection
Deliverables
• Design Plan • Market Analysis • Solution Comparisons • Process Flow Diagram • Material Balances
Team Members
Ahmad Abusubaih Emir Asani Xuan Liu Michael Simon

Design Plan

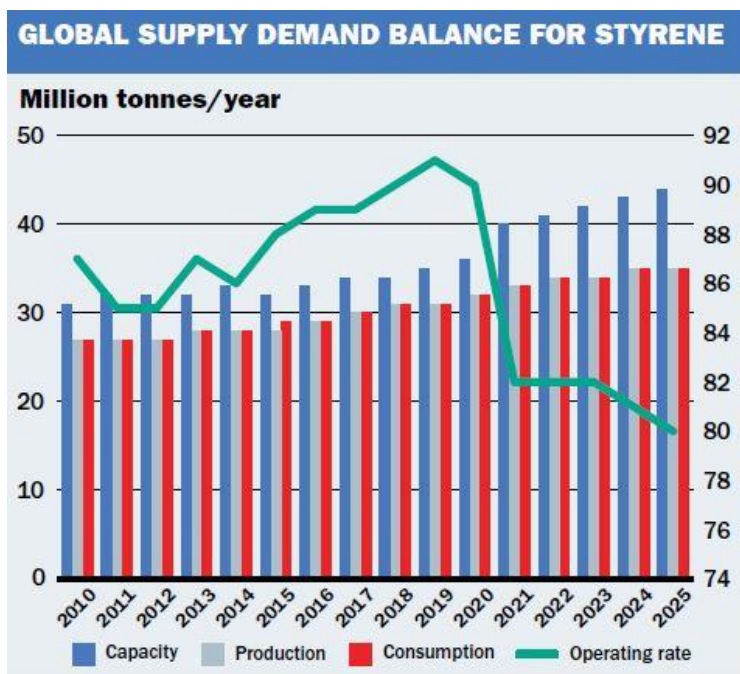
Making Styrene from Scratch Schedule & Plan				
Task	Assignee	Start Date	End Date	Duration (Days)
First Meeting - Set Goals	All	9-Sep	9-Sep	1
Preliminary Research	All	9-Sep	13-Sep	1
Second Meeting - Selected Solution	All	10-Sep	10-Sep	1
Alternative Soltuions Research	Ahmad & Xuan	10-Sep	13-Sep	4
Process Research	Emir & Michael	10-Sep	13-Sep	4
Third Meeting - Planning Execution	All	13-Sep	13-Sep	1
Design Plan	Ahmad & Emir	13-Sep	16-Sep	4
Project Charter	Michael & Xuan	13-Sep	16-Sep	4
Introdcution and Objective	Ahmad & Emir	13-Sep	15-Sep	3
Importance and Customer Requirments	Ahmad & Xuan	13-Sep	14-Sep	3
Alternative Soltuions	Xuan	13-Sep	15-Sep	3
Chosen Solution	Ahmad & Xuan	13-Sep	16-Sep	4
Design/Assumptions	Emir & Xuan	13-Sep	16-Sep	4
Market Analysis	Ahmad & Michael	13-Sep	15-Sep	3
Material Balance	Emir & Michael	12-Sep	14-Sep	3
PFD and BFD Design	Emir & Michael	10-Sep	14-Sep	5
Report Compilation	All	14-Sep	17-Sep	4
Editing and Formatiing	All	16-Sep	17-Sep	2

Introduction

Objective

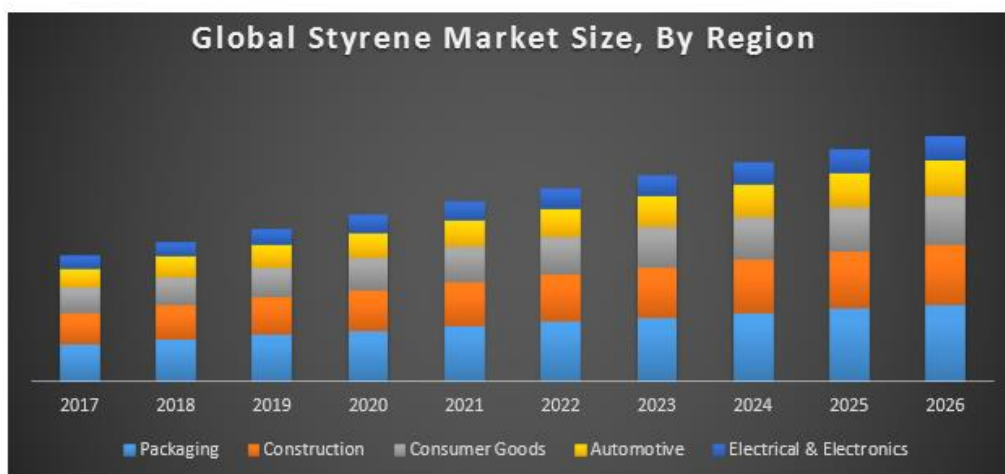
Styrene is an important component in the production of latex, synthetic rubber, and polystyrene resins. Polystyrene resins are widely used in many categories such as packaging, food service and appliances. Some items within those categories include disposable cups, foam insulation, and packaging peanuts [3]. Styrene is very important in our economy because it is long lasting and greatly cost-effective. The Independent Chemical Information Service estimates that the production of styrene will increase about five million tonnes/year in the next half-decade. As production and consumption of these items continue to increase, the demand of styrene also increases [4].

Figure 1: Styrene Supply and Demand (ICIS Chemical Business)



The best way to attract the high-demanding consumers, is to create the purest and low-costing styrene monomer. Therefore, the important aspects of having the most pure styrene results in an increase of strength, durability, and safety; those three characteristics of styrene are highly valued by the consumer, so the goal is to produce styrene monomer that is pure. As shown in the figure below, the global market value for styrene is estimated to increase by double its size by year 2026 [3]

Figure 2: Future Styrene Markets (Market Research PVT.)



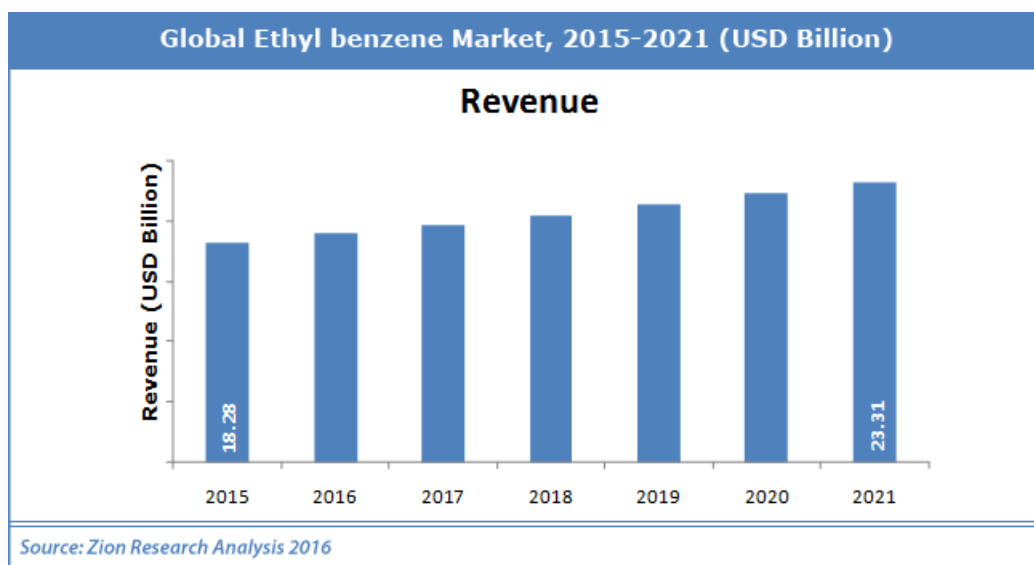
To reach our goal, our plan is set to produce 500 thousand metric tons per year to keep up with demand in the future. In order for the goal to be reached, production will begin with ethylene and benzene, rather than the traditional petrochemical way of starting with ethylbenzene since that is their byproduct, through dehydrogenation to create styrene.

Importance

Therefore, by producing ethylbenzene on site, it will increase the capital outcome of the styrene plant. It is more efficient to produce the ethylbenzene onsite, rather than in a separate plant from the styrene monomer. This will allow the plant to not only be a styrene producer, it can also be a supplier of ethylbenzene to other styrene plants. By doing so, this will increase market value and attract more consumers [6]

As shown in the figure below the market value for styrene has increased over the years and has predicted to continue raise in the near future. In the year 2015, the revenue was at 18.28 billion dollars while it is estimated to increase to 23.31 billion dollars [2].

Figure 3: Global Ethylbenzene Market



Customer requirements

The best way to attract the high-demanding consumers, is to create the purest and low costing styrene monomer. Therefore, the goal is to produce styrene monomer that is pure and cost-efficient for the consumers. Consumers use styrene for many materials such as plastic, cups, hoses, carpeting, tires and much more. Once other companies learn about this cost-efficient process, it will attract them to request more supply of styrene monomer for their own production [3].

As shown in figure 2, the global market value for styrene is estimated to increase by double its size by year 2026. The different market requirements by consumers would include items of packaging, constructions, consumer goods, automotive, electrical and electronics. The most widely requested item out of them is the packaging products to assist in protecting shipment items [3].

Problem

Styrene monomer is in many of the consumer products that are used on a daily basis. This monomer is used in plastics, fiberglass, rubber, latex, etc. In the petrochemical industry, 30 million tons of styrene monomer was in demand per year by 2016 [2]. Styrene is becoming more costly due to the rate of shipment of the ethylbenzene from different producers to the styrene plant. Since styrene is in high demand for many consumers, there should be a more efficient way to produce it. The way to produce styrene in cost efficient way is to create the ethylbenzene in the same plant as the styrene monomer. This will reduce the cost of shipment that would have been spent if ethylbenzene was purchased from other vendors. Also, since the process of

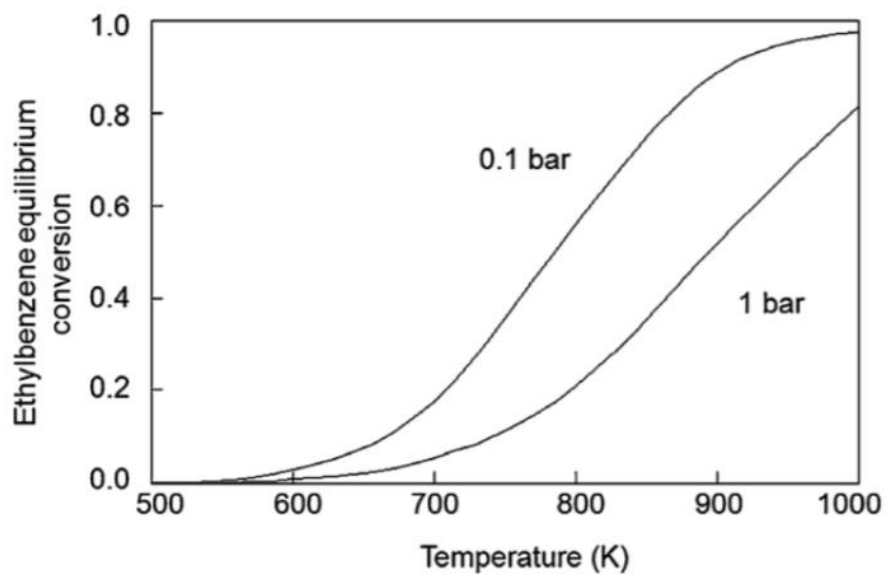
ethylbenzene exothermic, it provides the energy that is needed to produce styrene more efficiently [6].

The problem with the current way of creating styrene monomer is the high cost of obtaining ethylbenzene that is needed throughout this method. The cost includes the current cost of styrene, supplier profit and shipment cost. By eliminating the supplier profit and shipment cost, it will make the creation of styrene more cost-efficient. The way to start this cost-efficient process is to begin producing ethylbenzene in the same place as the styrene monomer is being produced [6].

Furthermore, the production of styrene requires high temperatures that would be more ideal to be absorbed from the formation of ethylbenzene. The high energy consumption of the process is due to the low conversion per pass [5]. This is from the high temperatures and the equilibrium limitations of the endothermic process. The figure below shows the effects on ethylbenzene equilibrium conversion from low pressure and high temperatures perspective. Ethylbenzene is produced by using energy at a temperature of 580 to 630 °C and is an exothermic reaction [5].

The process of ethylbenzene dehydrogenation requires the presence of steam and had 3 main roles. The first role is it lowers ethylbenzene partial pressures to minimize loss of thermal cracking which shifts the equilibrium towards styrene. The second role offers the necessary heat needed for the reaction. The third role is catalyst cleaning by reacting with carbon, producing hydrogen and carbon dioxide [1].

Figure 4: Ethylbenzene conversion vs. Temperature



Possible Solutions and Competing Processes

Styrene is one of the most important monomers for the polymer industry. As a result, with the continuous improvement and development of styrene production process by engineers, styrene now has many synthesis processes for chemical plants to choose. There are, ethylbenzene (EB)-based catalytic dehydrogenation technology, propylene oxide–styrene monomer (POSM) process, and selective oxidative dehydrogenation process [10].

Ethylbenzene (EB)-based catalytic dehydrogenation technology

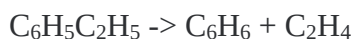
Ethylbenzene (EB)-based catalytic dehydrogenation technology refers to the high-temperature catalytic dehydrogenation of ethylbenzene in the presence of excess water vapor to produce styrene. This technology is mainly restricted by its catalysts. Typical operating conditions for catalytic dehydrogenation is following [10].

- The reaction temperature is 500 ~ 650 °C;
- The reaction pressure is minus 2 kg/cm²;
- Vapor/hydrocarbon (weight) ratio is (1~3):1.

The dehydrogenation of ethylbenzene to styrene is a reversible endothermic molecular reaction.



Each generation of 1 mol of styrene absorbs 125.4kJ. Therefore, high temperature and low pressure are favorable for the formation of styrene. Dehydrogenation of ethylbenzene to styrene, along with a series of side reactions, can produce benzene, toluene, a small amount of methane and olefins and tar. The main side effects are as follows.



The high temperature superheated steam is added to the ethylbenzene feed in a certain proportion. On the one hand, the addition of superheated steam provides the heat required for the reaction, while also reducing the partial pressure of the components. On the other hand, the water vapor reacts with the carbon deposit on the surface of the catalyst, causing the catalyst to regenerate at any time. The gas phase product produced by the dehydrogenation reactor is subjected to condensation and condensate separation to obtain a mixture containing styrene,

unreacted ethylbenzene, toluene and a small amount of tar. They are separated by a rectification system to obtain polymer grade styrene.

Propylene oxide–styrene monomer (POSM) process

Propylene oxide–styrene monomer (POSM) process is another important method for the industrial production of styrene monomer. This method has special significance in industrial production due to the production of styrene and another important organic chemical product, propylene oxide. In the POSM process, ethylbenzene is used as a raw material, and at a temperature of 130 to 160 °C and a pressure of 0.3 to 0.5 MPa, ethylbenzene is first oxidized by oxygen in a liquid phase reactor to form ethylbenzene peroxide. The resulting ethylbenzene peroxide was concentrated to 17% and then entered the epoxidation process. Under the reaction temperature of 110 °C and a pressure of 4.05 MPa, it undergoes epoxidation with propylene to form propylene oxide and alpha-methylbenzyl alcohol. Epoxy propane is obtained by distillation of epoxide reaction solution. Alpha-methylbenzyl alcohol was dehydrated to produce styrene at 260 °C and under normal pressure. The mass ratio of styrene to propylene oxide in the reaction product is 2.5:1. The whole process can be divided into main processes such as peroxidation, epoxidation, dehydration and hydrogenation [11].

Selective oxidative dehydrogenation process

In this process, hydrogen generated by dehydrogenation of ethylbenzene is selectively oxidatively burned to shift the reaction equilibrium in a direction favorable for the formation of styrene. At the same time, it also provides the heat required for the reaction, which can greatly increase the conversion rate of styrene production. This process achieves the same selectivity as

conventional dehydrogenation, but the production of styrene per unit mass requires 34% less steam than conventional processes. Its energy saving advantages are quite obvious [12].

The dehydrogenation of ethylbenzene is a reaction that absorbs heat and increases the number of molecules. High temperature and low pressure are favorable for the reaction. The traditional catalytic dehydrogenation method uses superheated steam as a heat carrier to provide the heat required for the reaction, and at the same time reduce the partial pressure of each component in the system to obtain a higher conversion rate. This process selects a different method than the conventional catalytic dehydrogenation process to selectively remove H_2 from the reaction to improve the equilibrium conditions of the reaction. The catalyst selectively oxidizes hydrogen in the presence of hydrocarbons and steam to alter the equilibrium of the system. The heat generated by this highly exothermic reaction can raise the temperature of the reaction mixture back to the level required for the dehydrogenation reaction. The use of this heat of reaction saves the superheated steam required for reheating between reactor stages. Selective dehydrogenation shifts the reaction equilibrium to styrene, thereby reducing side reactions and increasing the styrene conversion of ethylbenzene.

The main difference between this process and the traditional process is the reaction part. The separation part is basically the same as the traditional process.

The steam/hydrocarbon ratio is significantly reduced when this process has the same selectivity as conventional processes. If the original production is maintained, the amount of utility can be reduced. If the same comprehensive ethylbenzene feed is kept, the yield can be increased [12].

Competing processes for styrene

These three technologies have their advantages and limitations. Ethylbenzene (EB)-based catalytic dehydrogenation technology is the most widely used method. This is because the technology of catalytic dehydrogenation is mature, and the yield of styrene is higher. However, the reaction temperature of the method is higher, the amount of steam is larger, and the conversion rate of ethylbenzene is lower, resulting in a larger partial load and higher energy consumption [10]. If the price of energy rises, the application of this method will be affected.

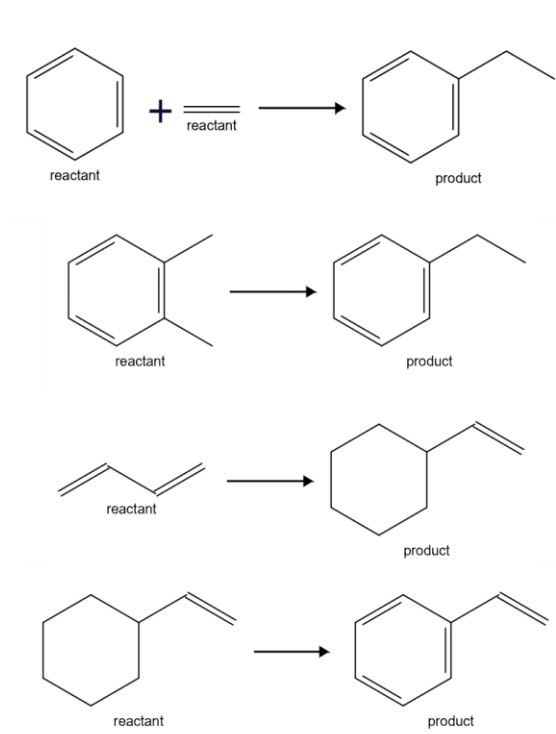
The propylene oxide–styrene monomer (POSM) process produces styrene and co-produces the important organic chemical propylene oxide. It is characterized by the fact that no high temperature reaction is required. The endothermic reaction of the endothermic dehydrogenation of ethylbenzene and the exotherm of propylene oxidation is combined to produce styrene at a lower temperature. This method not only saves energy, but also solves the environmental pollution problem of propylene oxide production by chlorohydrin [11].

The disadvantages are long process, high investment, high by-products and low yield of styrene. The choice of this method depends mainly on the market demand for these two products and the consideration of equipment investment. It is greatly affected by the market conditions of the joint products, and the reaction is complicated, the by-products are large, the investment is large, the consumption of ethylbenzene and the energy consumption of the equipment are higher than that of the ethylbenzene dehydrogenation process. However, from the perspective of co-oxidation of propylene oxide, it can avoid the environmental pollution caused by the chlorohydrin method.

The selective oxidative dehydrogenation process not only has low energy consumption, high yield of styrene, but also has a complicated process and low investment in equipment [12]. Therefore, this method is expected to become the main production process of styrene after catalytic dehydrogenation. Many old devices can also be modified by this method in order to obtain higher efficiency or increase production. However, the safety and reliability of the process have not been obtained. Widely verified. And in the process of transformation will also bring losses in production and other possible safety hazards.

For the production of ethylbenzene, it is mainly synthesized by the following three methods. They are formed by catalytic alkylation of benzene with ethylene, or by isomer separation and catalytic isomerization by mixed xylenes, or by a two-step reaction of 1,3-butadiene [13]. Their reaction equation is shown in Figure 5.

Figure 5: 3 reaction equation to produce ethylbenzene



From the perspective of source and cost of raw materials, benzene and ethylene are abundant in source and can be easily obtained from naphtha or liquefied petroleum gas. And their price is lower than that of xylene and 1,3-butadiene. Thus, the catalytic alkylation of benzene with ethylene to form ethylbenzene is the main way for industrial production of ethylbenzene.

The alkylation of benzene with olefins is based on a positive carbon ion mechanism. The active center of the acidic catalyst allows the olefin to form a positive carbon ion, thereby chemically reacting with the adsorbed benzene on the active surface of the catalyst to form an alkylbenzene. The alkylation of benzene and ethylene under acidic catalyst conditions produces ethylbenzene. The alkylation process of ethylene and benzene is divided into a liquid phase method and a gas phase method.

The traditional liquid phase method uses AlCl_3 as a catalyst. With the development of molecular sieve catalysts, the liquid phase method has been further developed. The alkylation reaction conditions in the liquid phase methods are, a temperature of 150 to 300 °C, a pressure of 2 to 6 MPa, and a benzene-alkene ratio of 2 to 15. Under high WHSV conditions, ethylene is generally not fully converted. The transalkylation reaction conditions are as follows: temperature 230 to 300 °C, pressure 2 to 6 MPa. The one-way ethylbenzene selectivity of the liquid phase alkylation process is generally between 90% and 95%, and the selectivity for ethylation is generally 98%[14].

For the gas phase method, the reaction temperature of the gas phase method is generally 35 to 450 °C, and the pressure is 0.2 to 2.0 MPa, the ratio of feed benzene is 4-20. Alkyl transfer

reaction conditions is, temperature 260 to 480 °C, pressure 0.4 to 3.5 MPa, a benzene-alkene ratio of 1 to 50. The mass fraction of xylene in the gas phase process is as high as 0.2% [15].

Competing processes for ethylbenzene

The two methods mentioned above each have their advantages and disadvantages. For ethylbenzene produced by gas phase process, it is accompanied by more by-product xylene. If the xylene content of ethylbenzene is high, it will affect the quality of styrene. And the activity of the catalyst and the selectivity of the reaction also needs to be improved.

The liquid phase alkylation is characterized by a very low content of xylene in the product ethylbenzene. Both the gas phase method and the liquid phase method face problems such as a reaction temperature, a pressure, a high benzene to ethylene molar ratio (bene ratio), and a large amount of by-product tar.

Market Analysis

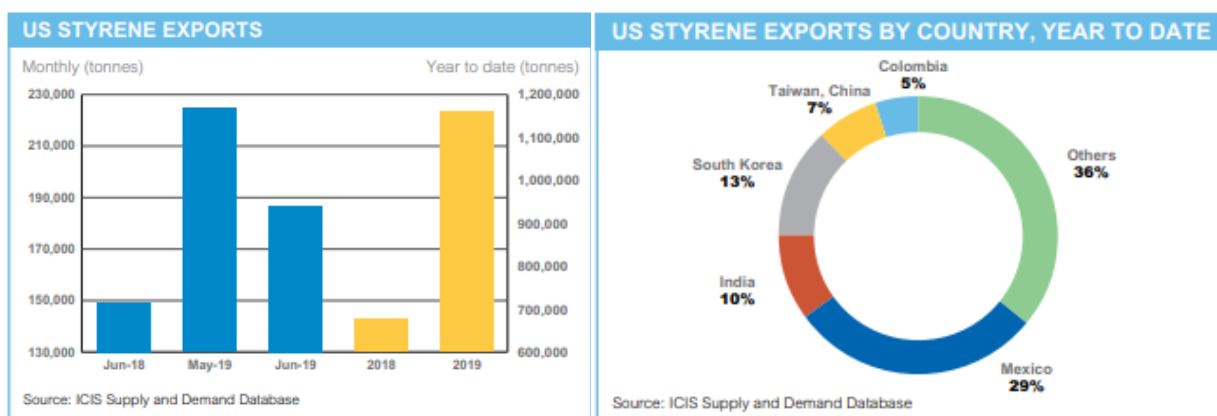
Styrene is almost exclusively used in the production of polymers, specifically polystyrene, expanded polystyrene, styrene butadiene latex, acrylonitrile butadiene styrene, styrene-butadiene rubber, and unsaturated polyester resins.[6] These styrene polymers and copolymers are used in many industries including food/beverage packaging, automotive manufacturing, tires, building insulation, paper, and household goods. Having several different downstream products and potential industries means that as long as there is an increasing demand for plastics, there will be a stable demand for styrene.

The United States and Saudi Arabia are both major exporters of styrene monomer, meaning that while the markets are saturated in these countries there is also a demand for styrene

in Asia, India, Mexico, and other styrene importing countries. With the US increasing its styrene exports by 25% from 2018,[8] building a new production plant would be most feasible in a country with growing styrene imports such as South Korea, Mexico, or Taiwan. Styrene projects in the United States are primarily focused on updating existing production sites for increased efficiency and competitiveness. Many sites are now making ethylene and benzene on site as well as ethylbenzene, although this is not crucial to profits due to the fact that these reagents are easily shipped by water or pipeline.[6]

Overall there are over 50 companies and 90 sites globally producing styrene monomer, but with a 30 million tons per year demand in 2016 [6] and a growing market there is still room for increased production. The main factors that need to be considered projects expanding production are transporting the raw materials or producing ethylbenzene on site, recycling energy from the ethylbenzene to the styrene reactor, and operating at a large enough scale offset capital costs.

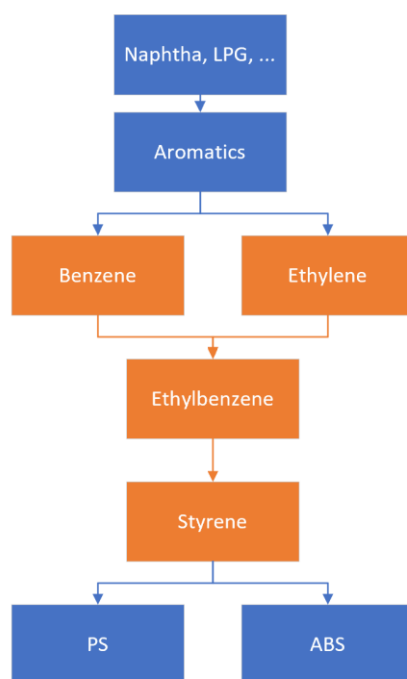
Figures 6 and 7: US Styrene Exports and Exports by Country [8]



Chosen Solution

The technology of catalytic dehydrogenation is mature, and the yield of styrene is higher than other ways. At the same time, benzene and ethylene are important preliminary processed products in the oil industry. Based on the information and pros and cons analysis presented in this proposal, it is the recommendation of this group that the ethylbenzene (EB)-based catalytic dehydrogenation technology and catalytic alkylation of benzene with ethylene technology be used as the method of conversion. The design ideas of our group are shown in Figure.

Figure 8: Production process diagram



Design Basis/Assumptions

Styrene is the simplest member of unsaturated aromatics. It is widely used as an important organic chemical for the production of styrene resins and styrene-butadiene rubber. In addition, styrene is one of the raw materials for the production of ion exchange resins and pharmaceutical products. Production of styrene monomer is extremely important in the chemical

industry as it is one of the main components in plastic materials, but larger demands of styrene come from the packaging and construction industries. Among the ethylene derivative products, styrene is second only to polyethylene, polyvinyl chloride and ethylene oxide. The production of styrene resin ranks third in the production of the world's top five synthetic materials, second only to polyethylene and polyvinyl chloride. According to statistics, the global benzene production capacity in 1999 was 21.698 Mt/a, and the demand in the same year was 20.02 Mt/a[16]. Even though the demand for styrene is so high, plant capacities are about 500 thousand tons per year [6]. Therefore, our goal is to produce about 50 lb/hr of styrene to reach our plant's fullest capacity. Then, we formed our basis around benzene and ethylene and assumed the compounds to be fed at a 1:1 ratio. For all our separations in our separators and distillation columns, we assumed a perfect separation, so no trace materials were leftover in any of the streams. As for all our parameters such as reaction conversions, temperatures and pressures, we selected those data from sources [7] and [17].

Material Balances

The styrene reactor material balances were carried out using ethylbenzene conversion of $C = 0.998$ and selectivities of $S_{C_6H_6} = 0.0249$, $S_{C_7H_8} = 0.0230$, $S_{C_8H_8} = 0.9520$. Conversions and selectivities were calculated from [7].

$$\dot{m}_{88,11} = \dot{m}_{88,19}$$

$$\dot{m}_{78,11} = \frac{\dot{m}_{88,11} \left(\frac{1}{\dot{m}_{88}} - 1 - \frac{\dot{m}_{66}}{1 - \dot{m}_{66}} \right)}{1 + \left(\frac{\dot{m}_{66}}{1 - \dot{m}_{66}} \right)}$$

$$\dot{m}_{66,11} = \frac{\dot{m}_{88,11}}{\dot{m}_{88}} - \dot{m}_{88,11} - \dot{m}_{78,11}$$

$$\dot{m}_{810,10} = \frac{\dot{m}_{88,11} + \dot{m}_{78,11} + \dot{m}_{66,11}}{\dot{m}}$$

$$\dot{m}_{810,11} = \dot{m}_{810,10} - \dot{m}_{88,11} - \dot{m}_{78,11} - \dot{m}_{66,11}$$

$$\dot{m}_{2,11} = \dot{m}_{88,11} + \dot{m}_{78,11}$$

$$\dot{m}_{4,11} = \dot{m}_{66,11}$$

For the ethylbenzene reactor material balances, the reactors were assumed to have conversions and selectivity of $C_1 = 0.735$, $C_2 = 0.97$, and $S_{C_8H_8} = 1.00$. Fresh ethylene and benzene were assumed to be fed at a 1:1 ratio, with an excess of 2:1 benzene to ethylene maintained by the recycle stream.

$$\dot{n}_{88,4} = \dot{n}_{88,10} - \dot{n}_{88,11}$$

$$\dot{n}_{24,2} = \frac{\dot{n}_{88,4}}{2}$$

$$\dot{n}_{24,4} = \dot{n}_{24,2} - \dot{n}_{88,4}$$

$$\dot{n}_{66,4} = \dot{n}_{24,1}$$

$$\dot{n}_{66,1} = \dot{n}_{24,1}$$

$$\dot{n}_{66,2} = \dot{n}_{66,1} + \dot{n}_{66,4}$$

$$\dot{n}_{24,3} = \dot{n}_{24,2} - \dot{n}_I(\dot{n}_{24,2})$$

$$\dot{n}_{66,3} = \dot{n}_{66,2} - (\dot{n}_{24,2} - \dot{n}_{24,3})$$

$$\dot{n}_{88,3} = \dot{n}_{24,2} - \dot{n}_{24,3}$$

After each reactor, the separation ethylbenzene and styrene were assumed to be perfect, approaching 100% purity, for more complete calculations thermodynamic calculations need to be made or simulated for actual separation efficiency.

Mass Balance Excel Sheet - All Units (kmol/hr)

Styrene Reactor			Settler				
Compound	In	Out	Compound	In	Out - S11	Out -S12	Out - S13
Ethylbenzene	52.344	0.105	Stryene	49.731	0.000	0.000	49.731
Stryene	0.000	49.731	Ethylbenzene	0.105	0.000	0.000	0.105
Hydrogen	0.000	50.937	Benzene	1.302	0.000	0.000	1.302
Benzene	0.000	1.302	Hydrogen	50.937	50.937	0.000	0.000
Toluene	0.000	1.207	Toluene	1.207	0.000	0.000	1.207
Methane	0.000	1.302	Methane	1.302	1.302	0.000	0.000
Steam	867.253	867.253	Steam	867.253	0.000	876.253	0.000
EB Separator 1			EB Separator 2				
Compound	In	Out - S4	Out -S5	Compound	In	Out - S6	Out -S7
Ethylene	0.157	0.000	0.157	Ethylene	0.157	0.000	0.157
Benzene	52.396	52.396	0.000	Ethylbenzene	52.239	52.239	0.000
Ethylbenzene	52.239	0.000	52.239				
Styrene Separator 1			Styrene Separator 2				
Compound	In	Out - S14	Out -S15	Compound	In	Out - S16	Out -S17
Stryene	49.731	0.000	49.731	Stryene	49.731	0.000	49.731
Ethylbenzene	0.105	0.000	0.105	Ethylbenzene	0.105	0.105	0.000
Benzene	1.302	1.302	0.000				
Toluene	1.207	1.207	0.000				
EB Reactor 1			EB Reactor 2				
Compound	In	Out	Compound	In	Out		
Ethylene	53.855	14.272	Ethylene	14.272	1.616		
Benzene	104.479	64.895	Benzene	64.895	52.239		
Ethylbenzene	0.000	39.583	Ethylbenzene	39.583	39.583		

Figure 9: Block Flow Diagram (BFD)

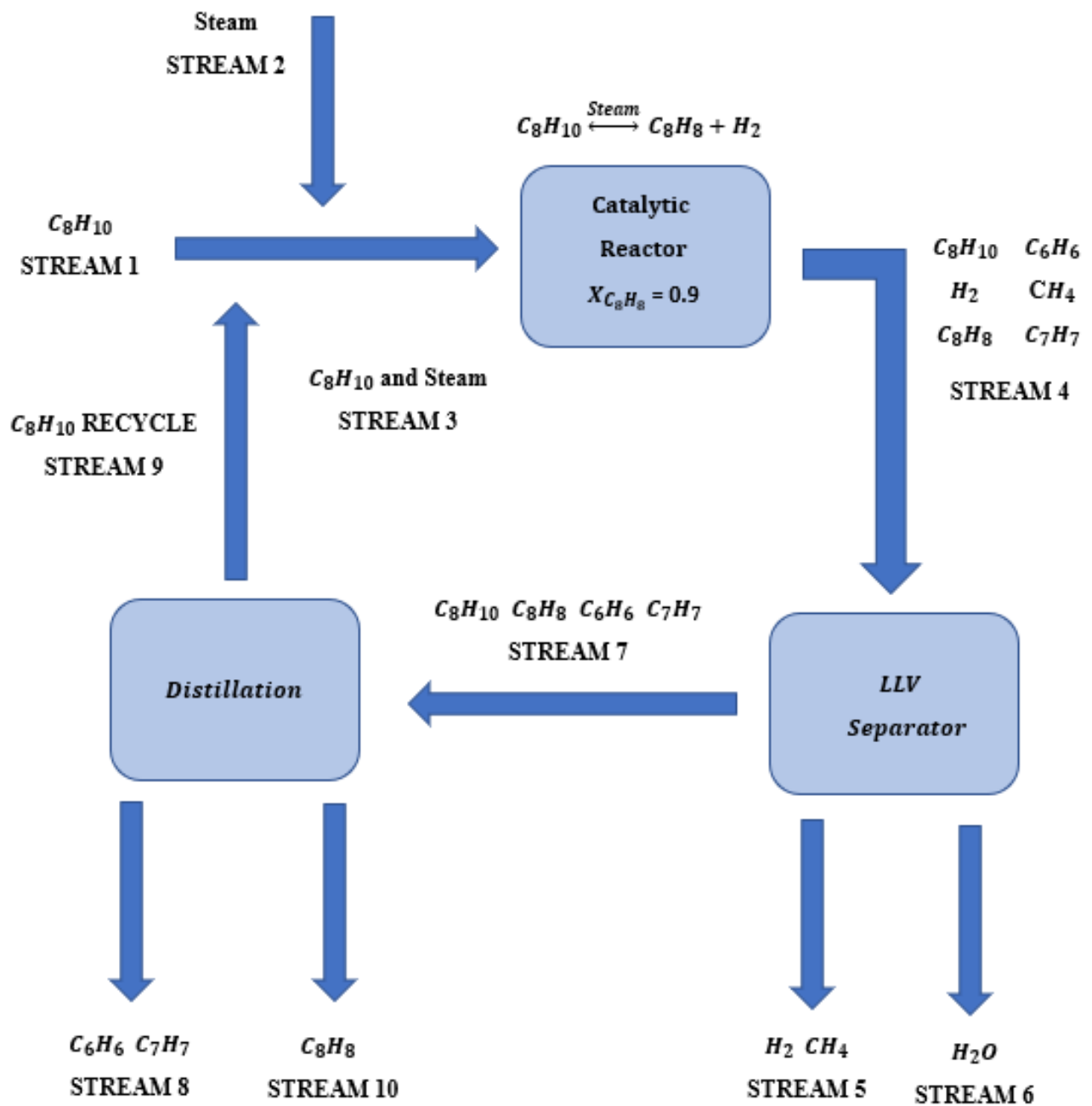
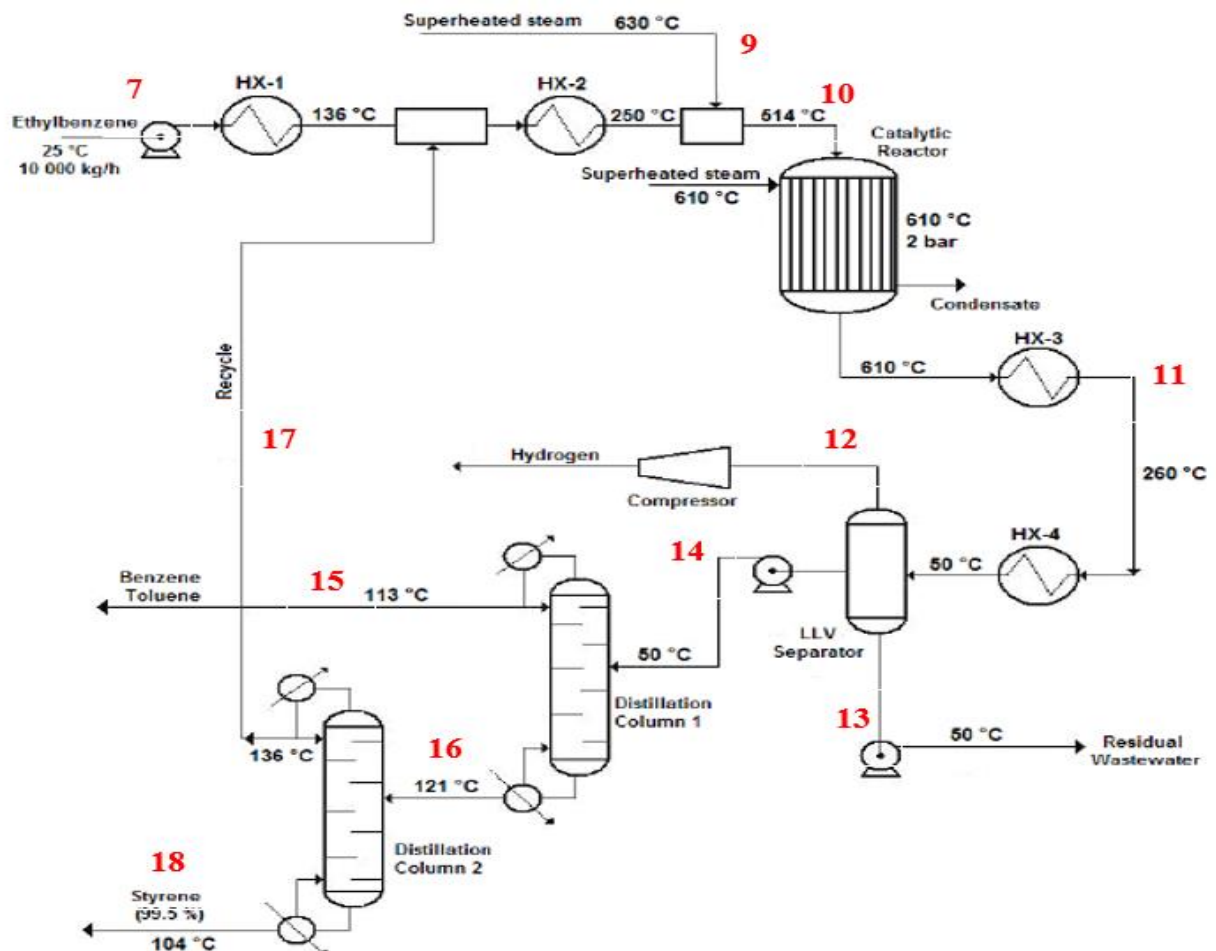
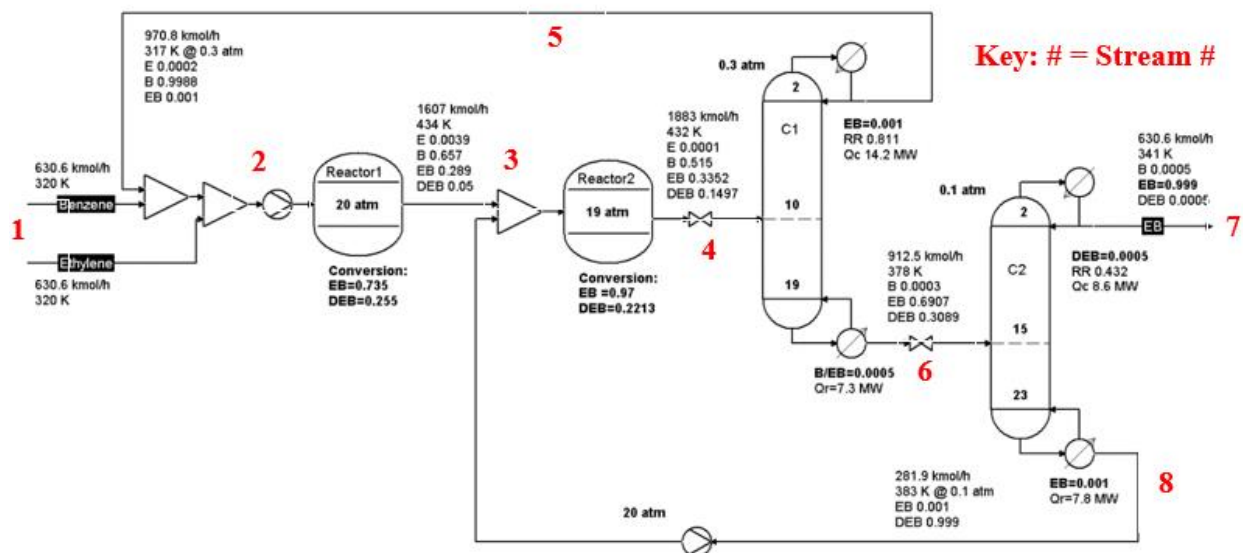


Figure 10: Process Flow Diagram [7] & [17]



Explanation and Purpose of Equipment

- 1) **Shell and Tube Heat Exchanger** - Shell and Tube Heat Exchangers - Used for heat transfer to cool down/heat up a fluid through another fluid's flow within its shell.
 - a) HX1 (Heat Exchanger 1) - Pre-heats the ethylbenzene to over 100°C for initial preparation for dehydrogenation.
 - b) HX2 (Heat Exchanger 2) - Increases temperature of ethylbenzene and Steam stream to completely vaporize at 250°C prior to reaction.
 - c) HX3 (Heat Exchanger 3) - After reaction, the mixture is cooled via cooling water within the heat exchanger.
 - d) HX4 (Heat Exchanger 4) - Further cools mixture via cooling water to about 50°C in preparation for separation.
- 2) **Distillation Column** - Fractionation column where it separates a mixture's components based on their boiling points.
 - a) C1 - Separates the benzene from the ethylbenzene/diethylbenzene mixture and recycles it back to the benzene feed stream.
 - b) C2 - Separates the diethylbenzene from ethylbenzene and then recycles the mixture back to the stream after the 1st reactor where diethylbenzene is initially present
 - c) Distillation Column 1 - Separates leftover benzene and toluene from the product mixture containing styrene.
 - d) Distillation Column 2 - Separates ethylbenzene from styrene mixture and recycles back to ethylbenzene feed stream.
 - e) Distillation Column 3 - Separates any trace components from the mixture to have a more pure styrene composition.
- 3) **Liquid - Liquid - Vapor (LLV) Separator** - A vessel where a multiphase mixture is separated into 3 different streams of light gas, waste, and useful mixtures.
 - a) (LLV Separator) - Separates the hydrogen gas and methane to the light gas stream, the waste water to the waste stream, and the rest of the components to the useful (middle stream).

4) **Reactor** - An apparatus where a system is controlled and is able to convert feed materials into different products.

- a) Reactor 1 - Reacts both feed streams of ethylene and benzene to form ethylbenzene and diethylbenzene
- b) Reactor 2 - diethylbenzene and benzene react to produce more ethylbenzene by benzene alkylation.
- c) Catalytic Reactor - Reacts ethylbenzene and steam to produce styrene and other byproducts.

5) **Compressor** - A machine that increases the pressure of fluid.

- a) Compressor - Increases pressure of the light gas to extract the light gas that's obtained from the LLV Separator.

6) **Pump** - a device that uses differentials of pressure to transport fluids

7) **Condenser** - a machine used as a cooling device; Placed at the top of all the distillation columns to cool down the hot vapors to liquid and send it back to the column to improve purity of product and assist with separation

8) **Reboiler** - a type of heat exchanger used to heat up liquid; placed at the bottom of all the distillation columns to boil the liquid to become vapor, and send back to the column to improve purity of product and assist with separation.

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